

The value associated with `favorite_language` ❶ contains extra whitespace at the end of the string. When you ask Python for this value in a terminal session, you can see the space at the end of the value ❷. When the `rstrip()` method acts on the variable `favorite_language` ❸, this extra space is removed. However, it is only removed temporarily. If you ask for the value of `favorite_language` again, the string looks the same as when it was entered, including the extra whitespace ❹.

To remove the whitespace from the string permanently, you have to associate the stripped value with the variable name:

```
>>> favorite_language = 'python '  
❶ >>> favorite_language = favorite_language.rstrip()  
>>> favorite_language  
'python'
```

To remove the whitespace from the string, you strip the whitespace from the right side of the string and then associate this new value with the original variable ❶. Changing a variable's value is done often in programming. This is how a variable's value can be updated as a program is executed or in response to user input.

You can also strip whitespace from the left side of a string using the `lstrip()` method, or from both sides at once using `strip()`:

```
❶ >>> favorite_language = ' python '  
❷ >>> favorite_language.rstrip()  
' python'  
❸ >>> favorite_language.lstrip()  
'python '  
❹ >>> favorite_language.strip()  
'python'
```

In this example, we start with a value that has whitespace at the beginning and the end ❶. We then remove the extra space from the right side ❷, from the left side ❸, and from both sides ❹. Experimenting with these stripping functions can help you become familiar with manipulating strings. In the real world, these stripping functions are used most often to clean up user input before it's stored in a program.

Removing Prefixes

When working with strings, another common task is to remove a prefix. Consider a URL with the common prefix `https://`. We want to remove this prefix, so we can focus on just the part of the URL that users need to enter into an address bar. Here's how to do that:

```
>>> nostarch_url = 'https://nostarch.com'  
>>> nostarch_url.removeprefix('https://')  
'nostarch.com'
```
